Optimal Endpoint Singular Factorization and the Coordinate-Matching Barrier

Majorization, Dynamical Outer Maps, and the Next All-Level Gate

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Abstract

RH-82 reduced the preferred Stage-A route to a factorization of the physical postblock state through the endpoint Gaussian resolution operator. This paper determines exactly what such a factorization requires and rules out an overly literal coordinate realization.

Let $\mathcal{R}$ have singular values $\rho_1 \geq \rho_2 \geq \cdots > 0$, and let a rank-$r$ target B_r have singular values $b_1 \geq \cdots \geq b_r > 0$. We prove the optimal two-sided factorization formula

$$\inf_{B_r = U\mathcal{R}V} \|U\| \, \|V\| = \max_{1 \leq j \leq r} \frac{b_j}{\rho_j}.$$

The lower bound is the ideal property of approximation numbers; equality is attained by aligning the two singular systems. For an arbitrary compact B, its truncated SVD therefore gives

$$B = U\mathcal{R}V + E_r, \qquad \|E_r\|_2 = \tau_r(B).$$

Consequently, singular-value majorization is sufficient for the RH-82 endpoint-to-postblock criterion. No coordinate alignment of singular vectors is needed.

A 192-bit audit compares the RH-16 linear endpoint Gram operator with all ten RH-77 frozen postblock states. At ranks within one of $\lceil H_\sigma \rceil + 2$, the optimal factor constant is at most 0.161572 and the largest Hilbert–Schmidt remainder is 1.232×10^{-9}. The largest relative remainder is 2.340×10^{-7}.

By contrast, direct coordinate projection onto sampled endpoint-row vectors leaves between 46.86% and 97.90% relative residual. Thus the identity outer map is decisively unsupported. The endpoint mechanism can survive only through a nontrivial dynamical rotation or, equivalently, an all-level singular-value majorization theorem.

The latter theorem remains open, so uniform Stage A1 and unconditional Stage A4 are not claimed. No Hilbert–Polya or Riemann-hypothesis conclusion is made.

Keywords: singular-value majorization; operator factorization; endpoint resolution; approximation numbers; effective rank.

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1 Why the outer maps matter

RH-82 proved exponential Hilbert–Schmidt tails beyond the half-logarithmic endpoint rank clock and showed that a bounded factorization

$$B_\sigma = U_\sigma \mathcal{R}_\sigma V_\sigma + E_\sigma \tag{1}$$

would supply the RH-77 effective-rank premise [3]. The most literal attempt is to identify sampled endpoint rows directly with the physical state coordinates. That would set one outer map approximately equal to the identity.

There is no mathematical reason for this. The endpoint operator describes which singular scales survive Gaussian resolution, while the intervening nonnormal dynamics may rotate their singular vectors almost arbitrarily. The correct invariant question is therefore spectral: do the physical singular values fit underneath the endpoint singular staircase?

2 The optimal factorization constant

Let $\mathcal{R} : \mathcal{H}_1 \to \mathcal{H}_2$ and $B : \mathcal{K}_1 \to \mathcal{K}_2$ be compact operators. Write their singular values in decreasing order as $\rho_j = s_j(\mathcal{R})$ and $b_j = s_j(B)$.

Theorem 2.1 (Optimal endpoint singular factorization)**.** *Suppose B_r has rank r and $\rho_r > 0$. Then*

$$\boxed{\inf\{\|U\|\,\|V\| : B_r = U\mathcal{R}V\} = \max_{1\le j\le r} \frac{b_j}{\rho_j}.} \tag{2}$$

The infimum is attained by bounded operators U and V supported on the first r singular directions.

Proof. For every factorization $B_r = U\mathcal{R}V$, the ideal property of approximation numbers gives

$$b_j = s_j(B_r) \le \|U\|\,\|V\|\;\rho_j, \qquad 1 \le j \le r.$$

This proves the lower bound in (2) [1].

Choose singular systems

$$\mathcal{R}v_j = \rho_j u_j, \qquad B_r y_j = b_j x_j.$$

Let V map y_j isometrically to v_j and vanish on the orthogonal complement. Let $Uu_j = (b_j/\rho_j)x_j$ for $j \le r$ and let U vanish on the remaining singular directions. Then $B_r = U\mathcal{R}V$, $\|V\| = 1$, and $\|U\| = \max_{j\le r} b_j/\rho_j$, proving equality. □

Corollary 2.2 (Optimal approximate factorization)**.** *For arbitrary compact B, let B_r be its truncated rank-r SVD. If $\rho_r > 0$, then*

$$B = U\mathcal{R}V + E_r, \qquad \|E_r\|_2 = \tau_r(B) = \left(\sum_{j>r} b_j^2\right)^{1/2}, \tag{3}$$

and the factor constant for B_r is exactly $\max_{j\le r} b_j/\rho_j$.

Thus the endpoint-to-postblock gate separates into two scalar assertions:

$$\max_{j\le r_\sigma} \frac{s_j(B_\sigma)}{s_j(\mathcal{R}_\sigma)} \le \operatorname{polylog}(1/\sigma), \tag{4}$$

$$\tau_{r_\sigma}(B_\sigma) \le \operatorname{polylog}(1/\sigma), \qquad r_\sigma = O(\log(1/\sigma)). \tag{5}$$

Together with the observability condition in RH-82, these bounds close the effective-rank corridor. They do not require a common coordinate basis.

3 Validated singular-majorization audit

For each archived noise scale, the audit constructs the linear endpoint Gram matrix from the exact powered-row affinity of RH-16 [2]. Its entries are propagated in 192-bit Arb arithmetic after exact lifting of the archived boundary clearances. If G_f is the binary64 Gram matrix and G_I its Arb enclosure, then

$$\|G_I - G_f\|_2 \le \|G_I - G_f\|_F.$$

Weyl's inequality therefore gives lower bounds for every endpoint singular value whose squared value exceeds the Gram defect.

The physical postblock state is recomputed in Arb from the exact frozen binary64 operator and source entries. Its binary64 SVD supplies candidate singular systems; the float-to-Arb state defect gives upper bounds for the physical singular values. Combining both sides certifies (2) at the resolved ranks.

σ	clock rank	factor rank	max factor	max remainder	min coord. residual
0.16	4	4	0.161572	1.232×10^{-9}	0.4686
0.08	5	5	0.021406	3.473×10^{-13}	0.6407
0.04	6	5	0.004181	3.564×10^{-10}	0.8657
0.02	6	6	0.006749	4.471×10^{-13}	0.9294
0.01	7	7	0.001145	8.276×10^{-14}	0.9372

Table 1: Outward-certified maxima over both channels, except the final column, which is the smaller coordinate residual lower bound. At $\sigma = 0.04$, the sixth endpoint Gram eigenvalue lies below the finite arithmetic resolution, so the certified factor rank is five.

The maximum factor constant occurs at the coarsest scale and is already well below one. The sole one-rank certification loss at $\sigma = 0.04$ increases the physical remainder only to 3.564×10^{-10}. No forbidden power of σ is visible at the anchors.

4 The coordinate-matching barrier

To test the identity outer-map shortcut, the audit samples the folded linear endpoint rows on the physical state grid, removes the limiting endpoint row, and projects B_σ onto the first $\lceil H_\sigma \rceil + 2$ left singular vectors of this coordinate dictionary. The state perturbation from binary64 to Arb is subtracted before forming the residual lower bound.

Every channel fails even a permissive 25% residual gate. The certified relative residuals range from 0.4687 to 0.9790. The mismatch grows toward small noise in the left channel.

Proposition 4.1 (Coordinate identity branch verdict)**.** *The archived clock-rank postblock states do not lie near the direct sampled endpoint-row coordinate subspaces. In particular, the numerical evidence does not support* (1) *with an identity left outer map and small remainder.*

This is intentionally a finite-scale branch verdict, not an impossibility theorem for all bounded outer maps. The optimal factorization theorem shows why: singular values match even when singular vectors are strongly rotated.

5 Next gate and claim boundary

RH-83 removes a potentially expensive and false obligation. One need not identify physical postblock singular vectors with endpoint row vectors. Instead, the next theorem should prove the scalar majorization (4) and tail bound (5) uniformly over the dyadic family. A natural route is to compare physical Gram eigenvalues with the exact endpoint Gram kernel through min–max or Schur-complement bounds, allowing nontrivial dynamical outer maps.

This paper proves the optimal factorization theorem and validates finite singular majorization at five scales. It does not prove all-level majorization, uniform Stage A1, unconditional Stage A4, a relative A5 determinant, a self-adjoint Hilbert–Polya operator, a $T \log T$ law, a prime-power trace formula, a zeta-zero identification, or the Riemann Hypothesis.

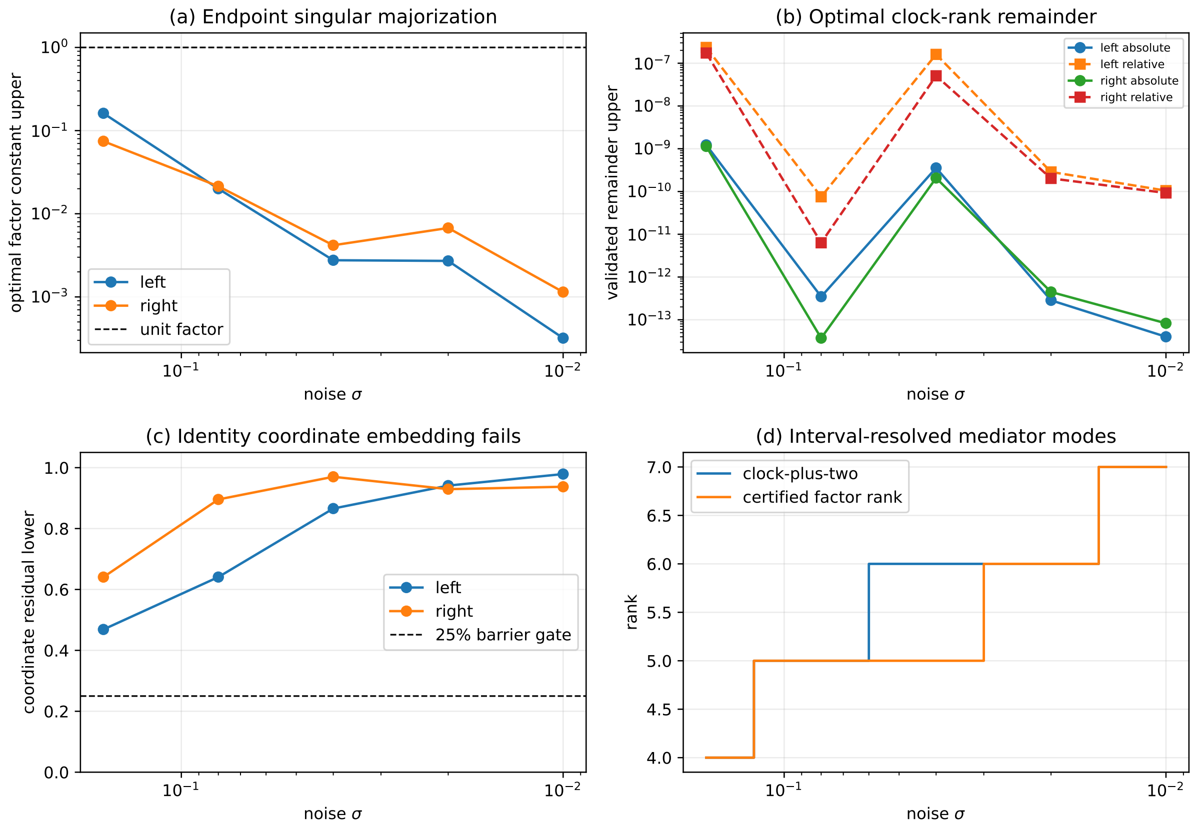

Figure 1: Optimal factor constants, SVD remainders, failure of direct coordinate matching, and interval-resolved endpoint ranks.

References

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